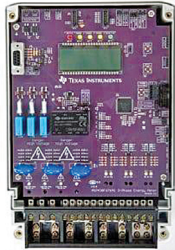


SMART ELECTRIC METERS

The author, Nidhi Agarwal, works as Technology Journalist at EFY

Class 0.5 3-Phase Smart Meter



This reference design for a Class 0.5 three-phase smart meter offers precise energy consumption measurement within $\pm 0.5\%$ in three-phase electrical systems. It enables accurate billing, energy management, and monitoring of active power and power factor.

The TIDM-THREEPHASEMETER-F67641, a reference design from Texas Instruments (TI), serves as an accurate three-phase electricity meter, providing energy consumption measurements within a range of $\pm 0.5\%$. This design enables accurate billing, streamlined energy management, and the monitoring of key parameters, including active power and power factor. The design ensures compliance with ANSI/IEC Class 0.5 accuracy requirements, adhering to the highest standards for precise energy measurement. Equipped with the TI Energy Library firmware, the MSP430F67641 system-on-chip (SoC) facilitates the calculation of all critical energy measurement parameters, including active and reactive power and energy, root mean square (RMS) current and voltage, power factor, line frequency, and fundamental and total harmonic distortion (THD) readings, providing a robust foundation for accurate energy analysis. The design also supports add-on communications modules for wireless connectivity such as ZigBee, Wi-Fi, Wireless M-Bus, and IEEE-802.15.4g, offering flexible data transmission and efficient communication with external systems. Additionally, the SoC incorporates a 160-segment display, presenting a convenient user interface for real-time energy consumption monitoring and enabling consumers to make informed decisions. Powering directly from the three-phase line voltage, the smart meter design simplifies the overall setup by eliminating the need for additional power supplies.

The reference design can be used for real-time monitoring of parameters like active power and power factor.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-class-0-5-3-phase-smart-meter>

Polyphase Smart Meter



This reference design for a polyphase smart meter is supported by various compliance certifications and facilitates metrology and calibration routines for billing. The design offers a dynamic range from 10mA to 100A, accommodating an expanded temperature spectrum.

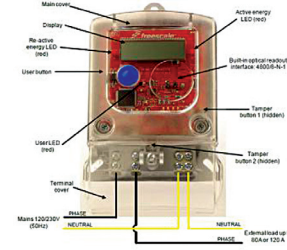
The YBILL-IT-RX21ASmartMeter is a reference design from Renesas for a polyphase smart meter, utilising the RX21A microcontroller unit (MCU) as its core component. The design is backed by various compliance certifications and facilitates metrology and calibration routines for billing. Depending on the specific configuration, the MCUs come with three, four, or seven independent 24-bit delta-sigma analogue-to-digital converters ($\Delta\Sigma$ -ADC), making the platform suitable for single-phase, split-phase, and three-phase plus neutral billing electricity meters. It has a dynamic range between 10mA and 100A over an enhanced temperature range and calibration data stored in the flash memory. It features all necessary Cenelec standards for electromagnetic compatibility (EMC) and safety, along with an intuitive PC GUI that provides engineers with comprehensive information during the design phase. The GUI enables engineers to trim and analyse data measured by the $\Delta\Sigma$ -ADC channels, offering graphical feedback on ADC data, as well as prepared data after calibration steps, frequency analysis, and separation of active/reactive energy usage. Renesas' 'Bill It! - smart meter RX21A' reference platform presents an all-in-one solution for engineers looking to develop polyphase meter designs. It integrates metrology and application processing capabilities, making it a powerful and efficient option for smart meter development.

The reference design can be used to design a customised billing electricity meter based on the user's needs.

Renesas

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-polyphase-smart-meter>

Single Phase Energy Meter



This reference design for a single-phase power meter, equipped with a high-performance analogue front end, sophisticated functionality, and sturdy components, ensures accurate energy measurement, compliance with international standards, and efficient operation. The design is tailored for measuring and recording active and reactive energy in single-phase two-wire networks with direct connection.

The Kinetis M Series one-phase power meter reference design from NXP offers an accurate, cost-effective, and feature-rich solution for energy metering in single-phase two-wire networks. The precise energy measurement, adherence to international standards and efficient operation are ensured by the high-performance analogue front end, advanced functionality, and durable components of the system. The reference design incorporates a high-performance analogue front end (AFE) and an embedded programmable gain amplifier (PGA), resulting in greatly improved energy measurement accuracy. Specifically tailored for single-phase two-wire networks with direct connections, the reference design efficiently measures and registers active and reactive energy. The design uses the Kinetis KM34Z128 microcontroller unit (MCU) with an ARM Cortex-M0+ core at up to 50MHz. An integrated buck converter ensures efficient power, with extra power for optional modules like non-volatile memory (NVM) for data logging. A 3-axis Xtrinsic tilt sensor is included for tamper detection. The design includes a battery for MCU backup during outages, ensuring uninterrupted meter operation for events like button presses or tampering. It features tamper switches for cover protection and an RF module for remote meter reading and monitoring. The Kinetis M Series design comes with extensive software, advanced metering algorithms for precision, and the MQX operating system with a peripheral library for easy application integration.

The reference design is suitable for residential metering, industrial metering, sub-metering, etc.

NXP

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